

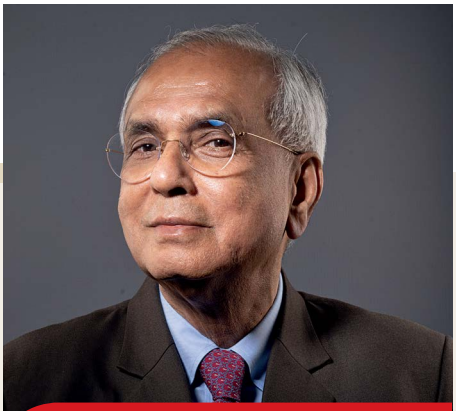
the supply, we also trigger the demand. And it's not just the question of demand generation. The concomitant challenges that are going to come in terms of the vast infrastructure requirement that production, transportation, storage of hydrogen will require... that itself is going to generate a lot of carbon emissions. So, the focus should be on creating new infrastructure, expanding the network and retrofitting the existing gas infrastructure. The supply chain ecosystem has to develop as fast as or faster than the production of electrolysers or the production of green hydrogen itself is concerned.

JAPAN'S NET-ZERO PLAN

Shinichiro Imahori: Japan is importing energy other than coal, but we have to achieve net-zero [emissions] by 2050. And in order to achieve this, gradual transition is required. That's why Japan is aiming to import green hydrogen. From Japan's perspective, we import energy from Australia mainly, and also from the US, Middle East and South Asia regions. But, backed by strong economic and political ties, India would be a very good candidate for [green hydrogen] suppliers and price.

MODELS OF COLLABORATION

Abhijeet Chatterjee: We are seeing a lot of large hydrogen projects being announced. We also need to look at how many of them are actually bankable. How many of them are actually reaching financial closure? For that, the demand really has to be sorted out.... Hitachi together with Marubeni is working on how an existing supply ecosystem can be used to develop a sustainable hydrogen ecosystem. We started with electrolysis using renewable energy to produce green hydrogen that was stored first in buffer tanks. And then we were able to design hydrogen cassettes for storage. Hitachi has also collaborated with Toyota for designing a hydrogen train, the first in Japan. That is something that we look to commercialise in Japan by 2030. So, these are some models of collaboration that can be useful for learning as we embark on the journey in India. ■



WHY JAPAN MATTERS
INDIA'S PERSPECTIVE

Rajiv Kumar
Former Vice-chairman, NITI Aayog

BRIDGING THE GAP

How to deepen cooperation between the two countries

Over the years, India and Japan have worked together to deepen trade and economic cooperation. Japan has played a pivotal role in building critical infrastructure in India. Former vice chairman of NITI Aayog **Rajiv Kumar** dug deeper to develop an understanding of the possibilities of making the relationship broader.

LACK OF REPRESENTATION

The representation of Japanese companies has grown, yet it's below par. I think they can come up much better and much bigger than what they've done till now. I especially refer to the fact that the businesses of Japan's Osaka region, the Kansai region, etc. are yet not found in India, and the supply chain has still not shifted.

THE LANGUAGE ISSUE

We had agreed to the fact that language presents a major block in our relationship. And there was a suggestion at one point in time to start offering the Japanese language to Indian high school children.... The bottleneck is the inability to communicate in the Japanese language and understand the culture. To the extent that we have to, if we want to work there. On their side, they get overwhelmed by India's diversity. They just don't know how to handle it... the very big differences in governance, in handling the ground reality, in handling the labour force, the skills etc. ■